

COUNTEREXAMPLES TO THE XU–HE–LU CONJECTURE ON PARTITE CYCLE SATURATION

ABSTRACT. Xu, He, and Lu conjectured that, for $\ell \geq 4$ and $n_1, n_2 \geq \ell + 2$,

$$\text{sat}(K_{n_1, n_2}, C_{2\ell}) = n_1 + n_2 + \ell^2 - 3\ell + 1.$$

For every $\ell \geq 4$, we construct a $C_{2\ell}$ -saturated spanning subgraph of $K_{2\ell-2, 2\ell-2}$ with $\ell^2 - 1$ edges. The conjectured value at these parameters is $\ell^2 + \ell - 3$, so the construction has $\ell - 2$ fewer edges than the conjectured value and refutes the conjecture. The smallest example is a 15-edge C_8 -saturated subgraph of $K_{6,6}$, where the conjecture predicts 17 edges.

1. RESULT

For graphs F and G , a spanning subgraph $H \subseteq G$ is *F-saturated relative to G* if H is F -free but $H + e$ contains a copy of F for every $e \in E(G) \setminus E(H)$. The minimum number of edges in such a graph is denoted by $\text{sat}(G, F)$.

Xu, He, and Lu [1, Theorem 1.1] proved that

$$(1) \quad \text{sat}(K_{n_1, n_2}, C_{2\ell}) \leq n_1 + n_2 + \ell^2 - 3\ell + 1$$

for $\ell \geq 3$ and $n_1, n_2 \geq \ell + 2$, and conjectured that this upper bound is the exact value. Verbatim, [1, Conjecture 6.2] states: “For $\ell \geq 4$ and $n_1, n_2 \geq \ell + 2$, $\text{sat}(K_{n_1, n_2}, C_{2\ell}) = n_1 + n_2 + \ell^2 - 3\ell + 1$.” For reference below we display the asserted equality,

$$(2) \quad \text{sat}(K_{n_1, n_2}, C_{2\ell}) = n_1 + n_2 + \ell^2 - 3\ell + 1 \quad (\ell \geq 4, n_1, n_2 \geq \ell + 2).$$

Theorem 1. *For every integer $\ell \geq 4$, there is a $C_{2\ell}$ -saturated spanning subgraph $H_\ell \subseteq K_{2\ell-2, 2\ell-2}$ with*

$$e(H_\ell) = \ell^2 - 1.$$

Consequently,

$$\text{sat}(K_{2\ell-2, 2\ell-2}, C_{2\ell}) \leq \ell^2 - 1,$$

and [1, Conjecture 6.2] is false for every $\ell \geq 4$.

Indeed, $2\ell - 2 \geq \ell + 2$, while the right-hand side of (2) at $n_1 = n_2 = 2\ell - 2$ is $\ell^2 + \ell - 3$, exceeding $\ell^2 - 1$ by $\ell - 2$. Thus Theorem 1 also improves the upper bound (1) by $\ell - 2$ edges at $n_1 = n_2 = 2\ell - 2$, which for $\ell = 4$ is the smallest admissible pair $n_1 = n_2 = \ell + 2 = 6$. It refutes (2) only at these balanced parameters; whether the equality in (2) holds when n_1 and n_2 are large relative to ℓ remains open.

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2. CONSTRUCTION AND PROOF

All path lengths below count edges. Fix $\ell \geq 4$ and put $r = \ell - 1$. Let

$$A_0 = \{a_0, a_1, \dots, a_{r-1}\}, \quad B_0 = \{b_0, b_1, \dots, b_{r-1}\},$$

and

$$X = \{x_1, \dots, x_r\}, \quad Y = \{y_1, \dots, y_r\}.$$

Take $A = A_0 \cup X$ and $B = B_0 \cup Y$ as the two partite classes. Let

$$R = K_{r,r} - a_0 b_0$$

on $A_0 \cup B_0$, and add the edges of the path

$$(3) \quad Q: a_0 - y_1 - x_1 - y_2 - x_2 - \dots - y_r - x_r - b_0.$$

Call the resulting graph H_ℓ . Its two parts have size $2r$, and

$$(4) \quad e(H_\ell) = (r^2 - 1) + (2r + 1) = r^2 + 2r = \ell^2 - 1.$$

For vertices u, v on Q , write $Q[u, v]$ for the unique u - v subpath of Q and $|Q[u, v]|$ for its length.

Lemma 2. *The graph R contains*

- (1) *an a_0 - b_0 path of every odd length $2s + 1$ with $1 \leq s \leq r - 1$;*
- (2) *for each $a \in A_0 \setminus \{a_0\}$, an a - a_0 path of every even length $2s$ with $1 \leq s \leq r - 1$.*

The symmetric statement in B_0 also holds.

Proof. For the first assertion, choose distinct $\alpha_1, \dots, \alpha_s \in A_0 \setminus \{a_0\}$ and distinct $\beta_1, \dots, \beta_s \in B_0 \setminus \{b_0\}$. Then

$$a_0 - \beta_1 - \alpha_1 - \beta_2 - \alpha_2 - \dots - \beta_s - \alpha_s - b_0$$

is a path of length $2s + 1$.

For the second, choose distinct $\beta_1, \dots, \beta_s \in B_0 \setminus \{b_0\}$ and distinct $\alpha_1, \dots, \alpha_{s-1} \in A_0 \setminus \{a_0, a\}$. The sequence of α_i is empty when $s = 1$, and

$$a - \beta_1 - \alpha_1 - \beta_2 - \dots - \alpha_{s-1} - \beta_s - a_0$$

is a path of length $2s$. The assertion in B_0 follows by symmetry. $\square$

Proof of Theorem 1. We first show that H_ℓ is $C_{2\ell}$ -free. A cycle contained in R has length at most $2r = 2\ell - 2$. Every internal vertex of Q has degree two and no neighbor outside Q . Hence a cycle meeting an internal vertex of Q must traverse all of Q and then return from b_0 to a_0 through R . Since $a_0 b_0 \notin E(R)$, such a core path has length at least three. The cycle therefore has length at least

$$(2r + 1) + 3 = 2r + 4 = 2\ell + 2.$$

Thus H_ℓ contains no $C_{2\ell}$.

It remains to show that the endpoints of every edge in $E(K_{2r,2r}) \setminus E(H_\ell)$ are joined in H_ℓ by a simple path of length $2r + 1 = 2\ell - 1$. Adding the missing edge then closes this path to a $C_{2\ell}$. Along Q we have

$$\begin{aligned} |Q[a_0, y_j]| &= 2j - 1, & |Q[y_j, b_0]| &= 2r - 2j + 2, \\ |Q[a_0, x_i]| &= 2i, & |Q[x_i, b_0]| &= 2r - 2i + 1. \end{aligned}$$

The missing edge a_0b_0 is witnessed by Q itself. Next consider a missing edge ay_j with $a \in A_0$. If $a \neq a_0$ and $j = 1$, concatenate the edge ab_0 with $Q[b_0, y_1]$; its length is $1 + 2r$. If $a \neq a_0$ and $j \geq 2$, Lemma 2 gives an a - a_0 path in R of length $2(r - j + 1)$, since $1 \leq r - j + 1 \leq r - 1$. Appending $Q[a_0, y_j]$ gives total length

$$2(r - j + 1) + (2j - 1) = 2r + 1.$$

If $a = a_0$ and $j \geq 2$, take an a_0 - b_0 path in R of length $2j - 1$ and append $Q[b_0, y_j]$; the total length is

$$(2j - 1) + (2r - 2j + 2) = 2r + 1.$$

Each concatenation is simple because its core portion meets Q only at the indicated endpoint or endpoints.

The map determined by

$$a_0 \leftrightarrow b_0, \quad a_k \leftrightarrow b_k \ (1 \leq k \leq r - 1), \quad x_i \leftrightarrow y_{r-i+1} \ (1 \leq i \leq r)$$

preserves R and reverses Q , and hence is an automorphism of H_ℓ . It maps the missing edges between A_0 and Y to those between X and B_0 , so the latter are handled as well.

Finally, consider a missing edge x_iy_j . The edges with $j = i$, and with $j = i + 1$ when $i < r$, already lie on Q . If $j \leq i - 1$, concatenate $Q[x_i, b_0]$, a b_0 - a_0 path in R of length $2(i - j) + 1$, and $Q[a_0, y_j]$. Lemma 2 applies because $1 \leq i - j \leq r - 1$, and the total length is

$$(2r - 2i + 1) + (2i - 2j + 1) + (2j - 1) = 2r + 1.$$

If $j \geq i + 2$, concatenate $Q[x_i, a_0]$, an a_0 - b_0 path in R of length $2(j - i) - 1$, and $Q[b_0, y_j]$. Here $1 \leq j - i - 1 \leq r - 2$, and the total length is

$$2i + (2j - 2i - 1) + (2r - 2j + 2) = 2r + 1.$$

In both cases the two subpaths of Q are vertex-disjoint, while the core path meets Q only at a_0 and b_0 ; hence the concatenation is simple. These cases exhaust all missing edges of the host graph, so H_ℓ is $C_{2\ell}$ -saturated. Equation (4) completes the proof. $\square$

For $\ell = 4$, the construction is $K_{3,3} - a_0b_0$ together with a seven-edge a_0 - b_0 path. It is a 15-edge C_8 -saturated spanning subgraph of $K_{6,6}$, whereas (2) gives 17.

REFERENCES

- [1] Y. Xu, Z. He, and M. Lu, *Partite saturation number of cycles*, Discrete Math. **349** (2026), no. 3, Paper No. 114802. doi:10.1016/j.disc.2025.114802; arXiv:2410.11194. The theorem and conjecture numbers cited here, and the quotation in Section 1, follow version 2 of the e-print.